

Zaichen Hao

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EDUCATION

University of Florida

Bachelor of Science in Computer Science and Mathematics

Gainesville, FL

Expected May 2028

UF Distinguished Scholars Scholarship Recipient | CLAS Dean's List, Spring 2026

Relevant Coursework: Operating Systems; Information Systems; Numerical Analysis; Software Development

Nanyang Technological University

Study Abroad Program in Machine Learning and Artificial Intelligence

Singapore

Jun 2025 - Aug 2025

PROJECTS

Low-Latency Order Matching Engine

Nov 2025

- Built a high-performance C++ order matching engine modeling core market microstructure with price-time priority, strict correctness guarantees, and integer-based price representation to avoid floating-point error.
- Optimized latency-critical data paths using ordered price-level maps and unordered ID-location mapping, achieving $\sim 2.2 \mu\text{s}$ effective per-order latency and $\sim 450\text{K}$ orders/sec on a single core.
- Profiled hot paths with Google Benchmark and Linux perf to identify bottlenecks and improve consistency.

Earnings Call Diffusion Model

Aug 2026

- Designed an end-to-end Python/pandas event-study pipeline that extracts market-relevant signals from earnings-call transcripts, aligns them with high-frequency market data, and models abnormal returns to estimate the speed of information incorporation into prices.
- Engineered event-level features for direction, surprise, novelty, and materiality, using historical-only novelty estimation to reduce look-ahead bias.
- Applied the pipeline to NVDA's Q2 FY2027 earnings-call dataset using official webcast captions and Databento quote data, extracting 189 candidate events and modeling 24 clustered shocks; estimated $T_{50}=50\text{s}$, $T_{90}=58\text{s}$, and $\lambda=0.0179$, ranking in the 95th percentile against placebo timing tests.

Film Revenue Predictor

Jul 2025

- Led a 7-person AI course project at Nanyang Technological University, collaborating to define project scope and technical direction while building machine learning models with pandas and scikit-learn to predict film revenue from genre, budget, language, cast, and other 2019–2024 features.
- Improved random forest performance through k-fold cross-validation and hyperparameter tuning, increasing R^2 from 0.47 to 0.86.

WORK EXPERIENCE

Zhongguancun Institute of Artificial Intelligence

Beijing, China

AI Research Engineering Intern

May 2026 – Aug 2026

- Automated technical research workflows in Python using web scraping, API integration, data parsing, and metadata analysis to evaluate 150+ academic papers and technical resources for AI History Show, an interactive web exhibition for the 70th anniversary of the Dartmouth Conference.
- Analyzed 50+ open-source repositories using OScanner to assess code quality, commit patterns, contributor activity, dependency health, and potential AI-generated code risks.

TECHNICAL SKILLS & ACTIVITIES

Technical Skills: C++, Python, Java, SQL, pandas, scikit-learn, Data Structure&Algorithms, Statistical Modeling, Market Microstructure, Low-Latency Systems, Linux perf, Google Benchmark, Unit Testing, Git, Docker, NumPy

Training: Selected as 1 of 8 Official Members from 100+ applicants for QiShi Club's Low Latency Systems course; Delivered lectures on instrumentation and performance optimization in the electronic trading ecosystem.

Publication: "Comparison Between Different Qubit Designs for Quantum Computing." *IJHSR*, 2024